

Unit Conversion

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1. [March US 2023]

A scientist takes two samples of Earth's crust, each with a mass of 3240 grams. Sample *A* has density of 3.0 grams per cubic centimeter, and sample *B* has a density of 2.7 grams per cubic centimeter. How much greater is the volume, in cubic centimeters, of sample *B* than sample *A*?

2. [March US 2023]

The walls of a small apartment were covered using exactly 3 gallons of paint. The paint was spread uniformly over a total area of 960 square feet. What was the rate, in gallons per square foot, at which the paint was used?

A) $\frac{1}{960}$

B) $\frac{1}{320}$

C) 320

D) 960

3. [March US 2023]

The area of a rectangular region is increasing at a rate of 20 square feet per hour. Which of the following is closest to this rate in square meters per minute?
(Use 1 meter = 3.28 feet)

A) 0.03

B) 0.10

C) 1.09

D) 2.03

4. [June 2023]

The speed of a white-throated needletail, a type of bird, in flight was measured to be 49 miles per hour. What was the white-throated needletail's measured speed, in kilometers per hour?
(Use 1 mile = 1.6 kilometers)

A) 30.6

B) 47.4

C) 50.6

D) 78.4

5. [August US 2023]

An object traveled 30 miles in 2 hours. What was the average speed, in miles per hour, of the object?

6. [August US 2023]

An object's speed is increasing at a rate of 6.80 meters per second squared. What is this rate in miles per minute squared, rounded to the nearest tenth? (Use 1 mile=1,609 meters.)

7. [November 2023]

A raised garden is in the shape of a right rectangular prism. Its base has a width of 3 feet and a length of 27 feet, and it will be filled 24 inches high with topsoil. The total cost of the topsoil needed to fill the garden to this height is 198 dollars. What is the unit cost, in dollars, per cubic yard, for the topsoil? (1 yard= 3 feet; 1 foot = 12 inches)

8. [March 2024]

A certain town has an area of 3.99 square miles. What is the area, in **square yards**, of this town? (1 mile = 1,760 yards)

A) 441

B) 7,022

C) 776,341

D) 12,359,424

9. [May 2024]

The area of a rectangular region is increasing at a rate of 260 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet.)

A) 0.40

B) 1.32

C) 14.21

D) 24.17

10. [March 2024]

A remotely operated vehicle (ROV) is used for underwater ocean research. When the ROV is at a depth of 41 fathoms, what is the ROV's depth in feet? (1 fathom = 6 feet)

11. [May US 2024]

An object's speed is increasing at a rate of 3.9 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth?

(Use 1 mile = 1,609 meters.)

- A) 0.1 C) 104.6
B) 8.7 D) 412.6

12. [August 2024]

A raised garden is in the shape of a right rectangular prism. Its base has a width of 3 feet and a length of 27 feet, and it will be filled 24 inches high with topsoil. The total cost of the topsoil needed to fill the garden to this height is 234 dollars. What is the unit cost, in dollars per **cubic yard**, for the topsoil?

(1 yard = 3 feet; 1 foot = 12 inches)



13. [August US 2024]

An object's speed is increasing at the rate of 12.1 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth?

(Use 1 mile=1,609 meters.)

- A) 0.5 C) 133
B) 27.1 D) 324.5

14. [November 2024]

An object's speed is increasing at a rate of 5.70 meter per second squared. What is this rate in miles per minute squared, rounded to the nearest tenth?

(Use 1 mile=1,609 meters.)



15. [October US 2024]

How many centimeters are equivalent to 35 meters?(1 meter = 100 centimeters)

16. [October US 2024]

An object's speed is increasing at a rate of 1.1 meters per second squared. What is this rate in miles per minutes squared, round to the nearest tenth?

(Use 1 mile = 1,609 meters.)

- A) 0 C) 29.5
B) 2.5 D) 1462.7

18. [December 2024]

An object moves at a speed of $\frac{3}{50}$ feet per second. What is this speed, in yards per second? (3 feet = 1 yard)

- A) 50 C) $\frac{9}{50}$
B) 6 D) $\frac{1}{50}$

17. [December US 2024]

The area of a rectangular region is increasing at a rate of 280 square feet per hour. Which of the following is closest to this rate in square meters per minute?

(Use 1 meter = 3.28 feet.)

- A) 0.43 C) 15.31
B) 1.42 D) 26.03

19. [March US 2025]

The table shows the distribution of people in a certain city by age group.

Age group	Proportion
Less than 18 years old	34%
18–40 years old	23%
41–65 years old	22%
Greater than 65 years old	21%

If a person in this city is selected at random, which of the following is closest to the probability of selecting a person who is greater than 65 years old, given that the person is at least 18 years old?

- A) 0.21 C) 0.45
B) 0.32 D) 0.62

20. [May 2025]

A certain hybrid car uses 39 gallons of gasoline to travel 1,950 miles. At this rate, how many kilometers does the car travel using 39 gallons of gasoline? (Use 1 mile = 1.6 kilometers.)

- A) 312 C) 31,200
B) 3,120 D) 312,000

21. [May 2025]

The bone mineral density of a certain person's thoracic spine was estimated to be 0.812 grams per square centimeter. What is this estimated bone mineral density in grams per square millimeter? (1 centimeter=10 millimeters)

- A) 81.2 C) 0.0812
B) 8.12 D) 0.00812

22. [June 2025]

An object's speed, in meters per hour, is increasing at a rate of 1,490 meters per hour per second. Which of the following is closest to this rate in feet per second squared? (Use 1 meter = 3.28 feet.)

- A) 0.126 C) 7.57
B) 1.36 D) 81.5

23. [June US 2025]

A container has a volume of 0.84 cubic meters. Which of the following is closest to this volume, in quarts? (Note: 100.00 centimeters = 1 meter. Use 946.35 cubic centimeters = 1.00 quart.)

- A) 0.09 C) 794.93
B) 7.95 D) 887.62

24. [June US 2025]

How many centimeters are equivalent to 47 meters? (1 meter = 100 centimeters)

25. [June US 2025]

An area of 56.00 square nautical miles is equivalent to k square kilometers. To the nearest tenth, what is the value of k ? (1 nautical mile = 1.852 kilometers)